A F M Saif

Research Engineer · LLM Post-Training
Reinforcement Learning · Distributed Training · Model Evaluation

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Profile

Machine learning researcher and engineer working on **LLM post-training, reinforcement learning, and distributed training**. Built SFT and **GRPO** post-training pipelines from scratch for 8–9B parameter models on multi-GPU clusters, and developed methods for parameter-efficient adaptation, model merging, and large-scale evaluation. Published at **NeurIPS, ICML, Interspeech, ICASSP, and IEEE/ACM TASLP**; two US patent applications with IBM. Open to relocation to San Francisco, New York, or Seattle.

Technical Skills

Post-Training Supervised fine-tuning, **GRPO** (rollout generation, group-relative advantage estimation, KL regularization to a reference policy), reward design, model merging, parameter-efficient fine-tuning (LoRA/PEFT), mixture-of-experts

Systems Distributed multi-GPU training, **FSDP** sharding, **DeepSpeed ZeRO**, memory-constrained training, checkpointing, CUDA, Slurm, Linux, Bash, Git

Frameworks **PyTorch** (custom training loops), Hugging Face Transformers, DeepSpeed, Weights & Biases, NumPy, SciPy

Evaluation Benchmark evaluation across mathematical reasoning, code generation, multilingual, instruction-following, and safety tasks; LLM-as-a-judge; inference and decoding pipelines

Optimization Bilevel, multi-objective, stochastic, and nonconvex optimization

Programming Python

Industry Experience

Summer 2026 **Research Extern – LLM Post-Training**, *IBM Research*, Yorktown Heights, NY

- Reproduced **LLM post-training pipelines from scratch** for **Llama-3.1-8B** and **Gemma-2-9B**, writing the training loop directly in PyTorch rather than using off-the-shelf trainers, and validated task-specialized models against reported results across mathematical reasoning, coding, multilingual, instruction-following, and safety benchmarks.
- Implemented **SFT and gradient-based GRPO** for mathematical reasoning on **NuminaMath-TIR** (72K examples), covering rollout generation, group-relative advantage estimation, and KL regularization against a reference policy.
- Trained 8–9B parameter models on **4×A100 GPUs**, resolving out-of-memory limits with **FSDP** sharding and **DeepSpeed**, and built end-to-end workflows spanning dataset preprocessing, distributed training, checkpointing, inference, and benchmark evaluation.
- Diagnosed **NaN training loss** caused by long mathematical prompts exceeding the context window: truncation removed the supervised target span, leaving affected batches with no valid loss tokens. Identified and filtered the affected examples to restore stable training.
- Implemented and evaluated **model-merging methods** — Task Arithmetic, TIES, DARE, Fisher Merging, RegMean, and Localize-and-Stitch — analyzing capability retention and task interference in multi-task models.
- Investigated **sparse parameter adaptation** for efficiently combining independently post-trained models while reducing cross-task interference.

Research Experience

2026 – Present **Researcher – LLM Post-Training, RL, and Alignment**, *Rensselaer Polytechnic Institute*, Troy, NY

- Built **training-free GRPO** pipelines applying group-relative advantage estimation and verifier-based rewards to optimize an external experience library, evaluated on **HumanEval**, **AIME**, and **HotpotQA** with **Qwen2.5-7B/14B** and **DeepSeek-R1-Distill-Qwen-14B**.
- Developed experience-selection and credit-assignment methods based on **Shapley values**, cardinality-restricted Shapley, greedy optimization, retrieval, and **LLM-as-a-judge** to identify which retrieved experiences improve versus degrade downstream performance.
- Investigated **agentic AI alignment and memory safety**, including harmful-memory propagation and robustness of agents interacting with external or shared memory.

2025 – 2026 **Researcher – Parameter-Efficient LLM Adaptation**, *Rensselaer Polytechnic Institute / IBM Research*, Troy, NY

- Developed **BELLA**, a parameter-efficient framework combining a speech encoder, Q-Former bridge, and **Qwen2.5-7B** decoder for multilingual speech recognition.
- Designed **mixture-of-experts LoRA** adaptation with task-specific and shared experts, adapting a multi-billion-parameter LLM with roughly 72M trainable parameters.
- Trained with **DeepSpeed ZeRO-2** and distributed multi-GPU training across **8×A100 GPUs**, using staged routing and bilevel optimization to learn routing and adaptation parameters jointly.
- Built evaluation and decoding pipelines combining candidate generation and LLM-based rescoring.

2023 – 2025 **Graduate Research Assistant – Foundation Models and Optimization**, *Rensselaer Polytechnic Institute*, Troy, NY

- Developed training methods for large pretrained models using **bilevel optimization** (joint supervised and self-supervised training) and **conflict-aware multi-objective optimization** for multilingual multi-task modeling, yielding publications at ICASSP, TASLP, Interspeech, and NeurIPS and two IBM–RPI patent applications.
- Contributed to optimization methods for diffusion models and preference-guided multi-objective learning published at **ICML 2025** and **NeurIPS 2024**.

Education

2023-Present **Ph.D., Electrical, Computer, and Systems Engineering**, *Rensselaer Polytechnic Institute*, Troy, NY
Focus: machine learning, large language models, and optimization.

2023 – 2025 **M.S., Electrical, Computer, and Systems Engineering**, *Rensselaer Polytechnic Institute*, Troy, NY
Focus: machine learning, large language models, and optimization.

2013 – 2018 **B.Sc., Electrical and Electronic Engineering**, *Bangladesh University of Engineering and Technology*, Dhaka, Bangladesh

Selected Publications

NeurIPS 2025 **Saif, A F M**, Chen, L., Cui, X., Lu, S., Kingsbury, B., Chen, T. "Objective Soups: Multilingual Multi-Task Modeling for Speech Processing."

ICML 2025 Xiao, Q., Yuan, H., **Saif, A F M**, Liu, G., Kompella, R., Wang, M., Chen, T. "A First-order Generative Bilevel Optimization Framework for Diffusion Models."

NeurIPS 2024 Chen, L., **Saif, A F M**, Shen, Y., Chen, T. "FERERO: A Flexible Framework for Preference-Guided Multi-Objective Learning."

TASLP 2024 Cui, X., **Saif, A F M**, Lu, S., Chen, L., Chen, T., Kingsbury, B., Saon, G. "Bilevel Joint Unsupervised and Supervised Training for Automatic Speech Recognition."

Interspeech 2024 **Saif, A F M**, Chen, L., Cui, X., Lu, S., Kingsbury, B., Chen, T. "M^{2}ASR: Multilingual Multi-Task Automatic Speech Recognition via Multi-Objective Optimization."

ICASSP 2024 **Saif, A F M**, Cui, X., Shen, H., Lu, S., Kingsbury, B., Chen, T. "Joint Unsupervised and Supervised Training for Automatic Speech Recognition via Bilevel Optimization."

Preprints "Experience Is Not Always the Best Teacher: Identifying Harmful Memories in Agentic LLMs" (2026); "BELLA: Efficient Bilevel Learning with LoRA for Multilingual ASR"; "Heterogeneous Self-Supervised Acoustic Pre-Training with Local Constraints."

Earlier Three first-author papers on deep learning for medical imaging, 2019–2021 (*IEEE TAI*, *IEEE Access* ×2).

Patents

Cui, X., **Saif, A F M**, Chen, T. "Joint Unsupervised and Supervised Training for Automatic Speech Recognition." U.S. Patent Application No. 19/293,954. Jointly owned by IBM and Rensselaer Polytechnic Institute.

Cui, X., **Saif, A F M**, Chen, L., Lu, S., Chen, T., Kingsbury, B. "A Method for Multilingual Multi-task Acoustic Modeling Based on Conflict-Avoidance Multi-objective Optimization." U.S. Patent Application No. 18/731,228. Jointly owned by IBM and Rensselaer Polytechnic Institute.

Service

Reviewer NeurIPS; IEEE Transactions on Signal Processing; IEEE Signal Processing Letters; IEEE Journal of Selected Topics in Signal Processing; IEEE JBHI; IEEE Access; Asilomar; IEEE SAM.